Topic: Representing piecewise functions

Question: Find the Fourier series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} -1 & -L \leq x < 0 \\ 1 & 0 \leq x \leq L \end{cases}$$

Answer choices:

A $$f(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 + (-1)^{n+1}}{n} \sin\left(\frac{n\pi x}{L}\right)$$

B $$f(x) = \frac{L^2}{2} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \cos\left(\frac{n\pi x}{L}\right) + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 + (-1)^{n+1}}{n} \sin\left(\frac{n\pi x}{L}\right)$$

C $$f(x) = \frac{2L}{\pi^2} \sum_{n=1}^{\infty} \frac{1 + (-1)^{n+1}}{n} \sin\left(\frac{n\pi x}{L}\right)$$

D $$f(x) = \frac{L^2}{2} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{\pi^2} \sum_{n=1}^{\infty} \frac{1 + (-1)^{n+1}}{n} \sin\left(\frac{n\pi x}{L}\right)$$

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Solution: A

We'll need to calculate A_0, A_n, and B_n to find the Fourier series representation.

For A_0 we get

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\ dx$$

$$A_0 = \frac{1}{2L}\int_{-L}^{0} -1\ dx + \frac{1}{2L}\int_{0}^{L} 1\ dx$$

$$A_0 = -\frac{1}{2L}x\Big|_{-L}^{0} + \frac{1}{2L}x\Big|_{0}^{L}$$

$$A_0 = -\frac{1}{2L}(0) - \left(-\frac{1}{2L}(-L)\right) + \frac{1}{2L}L - \frac{1}{2L}(0)$$

$$A_0 = -\frac{1}{2} + \frac{1}{2}$$

$$A_0 = 0$$

And for A_n we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{1}{L}\int_{-L}^{0} -1\cos\left(\frac{n\pi x}{L}\right)\ dx + \frac{1}{L}\int_{0}^{L} 1\cos\left(\frac{n\pi x}{L}\right)\ dx$$

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$$A_n = -\frac{1}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} + \frac{1}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L}$$

$$A_n = -\frac{1}{n\pi}\sin\left(\frac{n\pi(0)}{L}\right) - \left(-\frac{1}{n\pi}\sin\left(\frac{n\pi(-L)}{L}\right)\right) + \frac{1}{n\pi}\sin\left(\frac{n\pi L}{L}\right) - \left(\frac{1}{n\pi}\sin\left(\frac{n\pi(0)}{L}\right)\right)$$

$$A_n = \frac{1}{n\pi}\sin(-n\pi) + \frac{1}{n\pi}\sin(n\pi)$$

$$A_n = -\frac{1}{n\pi}\sin(n\pi) + \frac{1}{n\pi}\sin(n\pi)$$

$$A_n = 0$$

For B_n we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{1}{L}\int_{-L}^{0}\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{0}^{L}\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} - \frac{1}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L}$$

$$B_n = \frac{1}{n\pi}\cos\left(\frac{n\pi(0)}{L}\right) - \frac{1}{n\pi}\cos\left(-\frac{n\pi L}{L}\right) - \left[\frac{1}{n\pi}\cos\left(\frac{n\pi L}{L}\right) - \frac{1}{n\pi}\cos\left(\frac{n\pi(0)}{L}\right)\right]$$

$$B_n = \frac{1}{n\pi} - \frac{1}{n\pi}\cos(-n\pi) - \frac{1}{n\pi}\cos(n\pi) + \frac{1}{n\pi}$$

$$B_n = \frac{2}{n\pi} - \frac{1}{n\pi}\cos(n\pi) - \frac{1}{n\pi}\cos(n\pi)$$

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$$B_n = \frac{2}{n\pi} - \frac{2}{n\pi}\cos(n\pi)$$

$$B_n = \frac{2}{n\pi}(1 + (-1)^{n+1})$$

Then the Fourier series representation of the piecewise function on $-L \leq x \leq L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{1 + (-1)^{n+1}}{n} \sin\left(\frac{n\pi x}{L}\right)$$

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Topic: Representing piecewise functions

Question: Find the Fourier series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} -2x & -L \leq x < 0 \\ 2x & 0 \leq x \leq L \end{cases}$$

Answer choices:

A $$f(x) = 2L + \frac{8L}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

B $$f(x) = 2L + \frac{8L}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos\left(\frac{n\pi x}{L}\right) + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n (L+4)}{n} \sin\left(\frac{n\pi x}{L}\right)$$

C $$f(x) = L + \frac{4L}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

D $$f(x) = L + \frac{4L}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos\left(\frac{n\pi x}{L}\right) + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n (L+2)}{n} \sin\left(\frac{n\pi x}{L}\right)$$

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Solution: C

We'll need to calculate A_0, A_n, and B_n to find the Fourier series representation.

For A_0 we get

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\ dx$$

$$A_0 = \frac{1}{2L}\int_{-L}^{0} -2x\ dx + \frac{1}{2L}\int_{0}^{L} 2x\ dx$$

$$A_0 = -\frac{1}{2L}x^2\Big|_{-L}^{0} + \frac{1}{2L}x^2\Big|_{0}^{L}$$

$$A_0 = -\frac{1}{2L}(0)^2 - \left(-\frac{1}{2L}(-L)^2\right) + \frac{1}{2L}L^2 - \frac{1}{2L}(0)^2$$

$$A_0 = \frac{1}{2L}L^2 + \frac{1}{2L}L^2$$

$$A_0 = L$$

And for A_n we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{1}{L}\int_{-L}^{0} -2x\cos\left(\frac{n\pi x}{L}\right)\ dx + \frac{1}{L}\int_{0}^{L} 2x\cos\left(\frac{n\pi x}{L}\right)\ dx$$

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$$A_n = -\frac{2}{L}\int_{-L}^{0} x\cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$ to evaluate the first integral.

$$A_n = -\frac{2}{L}\left[\frac{Lx}{n\pi}\sin\left(\frac{n\pi x}{L}\right) - \frac{L}{n\pi}\int \sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_{-L}^{0} + \frac{2}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = -\frac{2x}{n\pi}\sin\left(\frac{n\pi x}{L}\right) - \frac{2L}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{0} + \frac{2}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = -\frac{2(0)}{n\pi}\sin\left(\frac{n\pi(0)}{L}\right) - \frac{2L}{(n\pi)^2}\cos\left(\frac{n\pi(0)}{L}\right)$$

$$-\left[-\frac{2(-L)}{n\pi}\sin\left(-\frac{n\pi L}{L}\right) - \frac{2L}{(n\pi)^2}\cos\left(-\frac{n\pi L}{L}\right)\right] + \frac{2}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = -\frac{2L}{(n\pi)^2} - \left(\frac{2L}{n\pi}\sin(-n\pi) - \frac{2L}{(n\pi)^2}\cos(-n\pi)\right) + \frac{2}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = -\frac{2L}{(n\pi)^2} + \frac{2L}{n\pi}\sin(n\pi) + \frac{2L}{(n\pi)^2}\cos(n\pi) + \frac{2}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2L}{(n\pi)^2}(-1)^n - \frac{2L}{(n\pi)^2} + \frac{2}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2L}{(n\pi)^2}(-1)^n - \frac{2L}{(n\pi)^2} + \frac{2}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

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Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$ to evaluate the second integral.

$$A_n = \frac{2L}{(n\pi)^2}(-1)^n - \frac{2L}{(n\pi)^2} + \frac{2}{L}\left[\frac{Lx}{n\pi}\sin\left(\frac{n\pi x}{L}\right) - \frac{L}{n\pi}\int \sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$A_n = \frac{2L}{(n\pi)^2}(-1)^n - \frac{2L}{(n\pi)^2} + \left[\frac{2x}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$A_n = \frac{2L}{(n\pi)^2}(-1)^n - \frac{2L}{(n\pi)^2} + \left[\frac{2L}{n\pi}\sin\left(\frac{n\pi L}{L}\right) + \frac{2L}{(n\pi)^2}\cos\left(\frac{n\pi L}{L}\right)\right]$$

$$-\left[\frac{2(0)}{n\pi}\sin\left(\frac{n\pi(0)}{L}\right) + \frac{2L}{(n\pi)^2}\cos\left(\frac{n\pi(0)}{L}\right)\right]$$

$$A_n = \frac{2L}{(n\pi)^2}(-1)^n - \frac{2L}{(n\pi)^2} + \frac{2L}{n\pi}\sin(n\pi) + \frac{2L}{(n\pi)^2}\cos(n\pi) - \frac{2L}{(n\pi)^2}$$

$$A_n = \frac{2L}{(n\pi)^2}(-1)^n - \frac{2L}{(n\pi)^2} + \frac{2L}{(n\pi)^2}(-1)^n - \frac{2L}{(n\pi)^2}$$

$$A_n = \frac{4L}{(n\pi)^2}(-1)^n - \frac{4L}{(n\pi)^2}$$

$$A_n = \frac{4L}{(n\pi)^2}((-1)^n - 1)$$

For B_n we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = \frac{1}{L}\int_{-L}^{0} -2x \sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{0}^{L} 2x \sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{2}{L}\int_{-L}^{0} x \sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_{0}^{L} x \sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ to evaluate the first integral.

$$B_n = \left[\frac{2x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{2}{n\pi}\int \cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_{-L}^{0} + \frac{2}{L}\int_{0}^{L} x \sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \left[\frac{2x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) - \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_{-L}^{0} + \frac{2}{L}\int_{0}^{L} x \sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \left[\frac{2(0)}{n\pi}\cos\left(\frac{n\pi(0)}{L}\right) - \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi(0)}{L}\right)\right]$$

$$-\left[\frac{2(-L)}{n\pi}\cos\left(-\frac{n\pi L}{L}\right) - \frac{2L}{(n\pi)^2}\sin\left(-\frac{n\pi L}{L}\right)\right] + \frac{2}{L}\int_{0}^{L} x \sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2L}{n\pi}\cos(-n\pi) + \frac{2L}{(n\pi)^2}\sin(-n\pi) + \frac{2}{L}\int_{0}^{L} x \sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2L}{n\pi}\cos(n\pi) - \frac{2L}{(n\pi)^2}\sin(n\pi) + \frac{2}{L}\int_{0}^{L} x \sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2L}{n\pi}(-1)^n + \frac{2}{L}\int_{0}^{L} x \sin\left(\frac{n\pi x}{L}\right)\,dx$$

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Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ to evaluate the first integral.

$$B_n = \frac{2L}{n\pi}(-1)^n + \left[-\frac{2x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2}{n\pi}\int \cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$B_n = \frac{2L}{n\pi}(-1)^n + \left[-\frac{2x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$B_n = \frac{2L}{n\pi}(-1)^n + \left[-\frac{2L}{n\pi}\cos\left(\frac{n\pi L}{L}\right) + \frac{2}{(n\pi)^2}\sin\left(\frac{n\pi L}{L}\right)\right]$$

$$-\left[-\frac{2(0)}{n\pi}\cos\left(\frac{n\pi(0)}{L}\right) + \frac{2}{(n\pi)^2}\sin\left(\frac{n\pi(0)}{L}\right)\right]$$

$$B_n = \frac{2L}{n\pi}(-1)^n - \frac{2L}{n\pi}\cos(n\pi) + \frac{2}{(n\pi)^2}\sin(n\pi)$$

$$B_n = \frac{2L}{n\pi}(-1)^n - \frac{2L}{n\pi}(-1)^n$$

$$B_n = 0$$

Then the Fourier series representation of the piecewise function on $-L \le x \le L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = L + \frac{4L}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n - 1}{n^2}\cos\left(\frac{n\pi x}{L}\right)$$

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Topic: Representing piecewise functions

Question: Find the Fourier series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} -\frac{L}{2} - x & -L \leq x < -\frac{L}{2} \\ 0 & -\frac{L}{2} \leq x \leq \frac{L}{2} \\ -\frac{L}{2} + x & \frac{L}{2} < x \leq L \end{cases}$$

Answer choices:

A $$f(x) = \frac{L}{8} + \sum_{n=1}^{\infty} \frac{2L}{(n\pi)^2} \left[(-1)^n - \cos\left(\frac{n\pi}{2}\right) \right] \cos\left(\frac{n\pi x}{L}\right)$$

B $$f(x) = \frac{L^2}{8} + \frac{2L}{\pi^2} \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n^2} - \frac{1}{n^2} \cos\left(\frac{n\pi}{2}\right) \right) \cos\left(\frac{n\pi x}{L}\right)$$

C $$f(x) = \frac{L^2}{8} - \frac{L}{\pi} \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n} + \frac{2}{n^2\pi} \sin\left(\frac{n\pi}{2}\right) \right) \sin\left(\frac{n\pi x}{L}\right)$$

$$+ \frac{2L}{\pi^2} \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n^2} - \frac{1}{n^2} \cos\left(\frac{n\pi}{2}\right) \right) \cos\left(\frac{n\pi x}{L}\right)$$

D $$f(x) = \frac{L^2}{8} - \frac{L}{\pi} \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n} + \frac{2}{n^2\pi} \sin\left(\frac{n\pi}{2}\right) \right) \cos\left(\frac{n\pi x}{L}\right)$$

$$+ \frac{2L}{\pi^2} \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n^2} - \frac{1}{n^2} \cos\left(\frac{n\pi}{2}\right) \right) \cos\left(\frac{n\pi x}{L}\right)$$

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Solution: A

We see that $f(-x) = f(x)$,

$$f(-x) = \begin{cases} -\frac{L}{2} - x & -L < x \leq -\frac{L}{2} \\ 0 & -\frac{L}{2} \leq x \leq \frac{L}{2} \\ -\frac{L}{2} + x & \frac{L}{2} \leq x < L \end{cases} = f(x)$$

so the function $f(x)$ is even, which means $B_n = 0$.

For A_0 we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\int_0^{\frac{L}{2}} 0\ dx + \frac{1}{L}\int_{\frac{L}{2}}^L \left(-\frac{L}{2} + x\right)\ dx$$

$$A_0 = -\frac{1}{L}\int_{\frac{L}{2}}^L \frac{L}{2} - x\ dx$$

$$A_0 - \frac{1}{L}\left(\frac{L}{2}x - \frac{1}{2}x^2\right)\Big|_{\frac{L}{2}}^L$$

$$A_0 = \frac{x^2}{2L} - \frac{x}{2}\Big|_{\frac{L}{2}}^L$$

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$$A_0 = \frac{L^2}{2L} - \frac{L}{2} - \left(\frac{\left(\frac{L}{2}\right)^2}{2L} - \frac{\frac{L}{2}}{2} \right)$$

$$A_0 = -\frac{L}{8} + \frac{L}{4}$$

$$A_0 = \frac{L}{8}$$

Then for A_n we get

$$A_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) \, dx$$

$$A_n = \frac{2}{L} \int_0^{\frac{L}{2}} 0 \cos\left(\frac{n\pi x}{L}\right) \, dx + \frac{2}{L} \int_{\frac{L}{2}}^L \left(-\frac{L}{2} + x\right) \cos\left(\frac{n\pi x}{L}\right) \, dx$$

$$A_n = -\int_{\frac{L}{2}}^L \cos\left(\frac{n\pi x}{L}\right) \, dx + \frac{2}{L} \int_{\frac{L}{2}}^L x \cos\left(\frac{n\pi x}{L}\right) \, dx$$

$$A_n = -\frac{L}{n\pi} \sin\left(\frac{n\pi x}{L}\right) \Big|_{\frac{L}{2}}^L + \frac{2}{L} \int_{\frac{L}{2}}^L x \cos\left(\frac{n\pi x}{L}\right) \, dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L) \, dx$, and $v = (L/n\pi)\sin(n\pi x/L)$ to evaluate the remaining integral.

$$A_n = \frac{2x - L}{n\pi} \sin\left(\frac{n\pi x}{L}\right) \Big|_{\frac{L}{2}}^L - \frac{2}{n\pi} \int_{\frac{L}{2}}^L \sin\left(\frac{n\pi x}{L}\right) \, dx$$

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$$A_n = \frac{2x - L}{n\pi} \sin\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2} \cos\left(\frac{n\pi x}{L}\right)\Bigg|_{\frac{L}{2}}^{L}$$

$$A_n = \frac{2L - L}{n\pi} \sin\left(\frac{n\pi L}{L}\right) + \frac{2L}{(n\pi)^2} \cos\left(\frac{n\pi L}{L}\right)$$

$$- \left[\frac{2\left(\frac{L}{2}\right) - L}{n\pi} \sin\left(\frac{n\pi\left(\frac{L}{2}\right)}{L}\right) + \frac{2L}{(n\pi)^2} \cos\left(\frac{n\pi\left(\frac{L}{2}\right)}{L}\right)\right]$$

$$A_n = \frac{L}{n\pi} \sin(n\pi) + \frac{2L}{(n\pi)^2} \cos(n\pi) - \frac{2L}{(n\pi)^2} \cos\left(\frac{n\pi}{2}\right)$$

$$A_n = \frac{L}{n\pi} \sin(n\pi) + \frac{2L}{(n\pi)^2} \cos(n\pi) - \frac{2L}{(n\pi)^2} \cos\left(\frac{n\pi}{2}\right)$$

$$A_n = + \frac{L}{n\pi}(0) + \frac{2L}{(n\pi)^2}(-1)^n - \frac{2L}{(n\pi)^2} \cos\left(\frac{n\pi}{2}\right)$$

$$A_n = \frac{2L}{(n\pi)^2}\left[(-1)^n - \cos\left(\frac{n\pi}{2}\right)\right]$$

Then the Fourier series representation of the piecewise function is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{L}{8} + \sum_{n=1}^{\infty} \frac{2L}{(n\pi)^2}\left[(-1)^n - \cos\left(\frac{n\pi}{2}\right)\right] \cos\left(\frac{n\pi x}{L}\right)$$

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